The Coulomb charge under refinement and why it cannot be renormalised

Illustrated guide, supplementary ideas, code verification

A companion document for editors and referees

Paper:	*Exact non-renormalisation of the Coulomb charge under refinement*
Journal:	Physical Review D (revtex4-2, two-column, 8 pages)
Verification:	5 Python scripts, full suite < 1 s, logs included
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In one sentence

The extraction of a continuum coupling from a discrete substrate is turned into a **theorem with an if-and-only-if** plus a **scoped no-go**; and the refinement flow is kept strictly apart from the renormalisation-group flow. No free parameters, no fits, everything recomputed in two seconds.

1 Four objects

Four things are routinely conflated in the literature, and once they are conflated every statement about "running" becomes ambiguous:

1. **The discrete charge**: defined by the divergence theorem, *exact*, no limit needed.
2. **The estimator**: the numerical rule that compares that charge with $1/r$. All bias lives here.
3. **The refinement flow**: a change of (mesh, estimator) at fixed physics, generated by $q_k \mapsto q_{k+1}$.
4. **The renormalisation-group flow**: a change of the observation scale μ, with the modes between μ and $\mu + d\mu$ integrated out.

The whole content of the paper compresses into two sentences: with the right estimator the third flow **vanishes identically**; and that fact **says nothing** about the fourth.

2 Theorem 1: the exactness criterion

Cut space into shells $[r_n, r_{n+1}]$ with arbitrary mesh ratio $q = r_{n+1}/r_n$. The discrete Gauss law reads

$$V_n - V_{n+1} = \frac{r_{n+1} - r_n}{\Omega_d \rho_n^{d-1}}.$$

Theorem 1

For every $n < N$ and every increasing sequence, $V_n - V_N = \Omega_d^{-1}(r_n^{2-d} - r_N^{2-d})$ holds **if and only if**

$$\rho_n^{d-1} = \frac{r_{n+1} - r_n}{r_n^{2-d} - r_{n+1}^{2-d}} \qquad (\forall n).$$

Proof. Telescoping sum, compared term by term; each term is positive and independent, so the condition is necessary shell by shell.

"Geometric mean of areas" is three-dimensional only

In $d = 3$ the condition is $\rho_n^2 = r_n r_{n+1}$, readable as the geometric mean of the two sphere *areas*. That is an accident of area $\propto r^2$. In $d = 4$ the condition is $\rho_n^3 = r_n^2 r_{n+1}^2/(r_n + r_{n+1})$, which is **not** an area mean; in general dimension only the displayed formula is correct.

3 Lemma 1: the exponents, checked item by item

Lemma 1 (Shell scaling)**.** *Assume (i) ρ_n satisfies Theorem 1; (ii) the mesh ratio q is the same on every shell; (iii) the source field is the Coulomb field of that dimension, $V \propto r^{2-d}$. Then for a trial potential r^{-p} the returned amplitude is*

$$A_n = \frac{1}{\Omega_d} \frac{r_n^{2-d} - r_{n+1}^{2-d}}{r_n^{-p} - r_{n+1}^{-p}} = \frac{r_n^{p-d+2}}{\Omega_d} \cdot \frac{q-1}{q(1-q^{-p})},$$

so $A_n \propto r_n^{p-d+2}$ exactly, and

$$\boxed{A_{n+1}/A_n = q^{p-d+2}}$$

independently of n. In three dimensions the exponent is $p-1$, which is where the form q^{p-1} belongs; marginality $p = d-2$ gives $q^0 = 1$ in every dimension.

Each hypothesis is load-bearing:

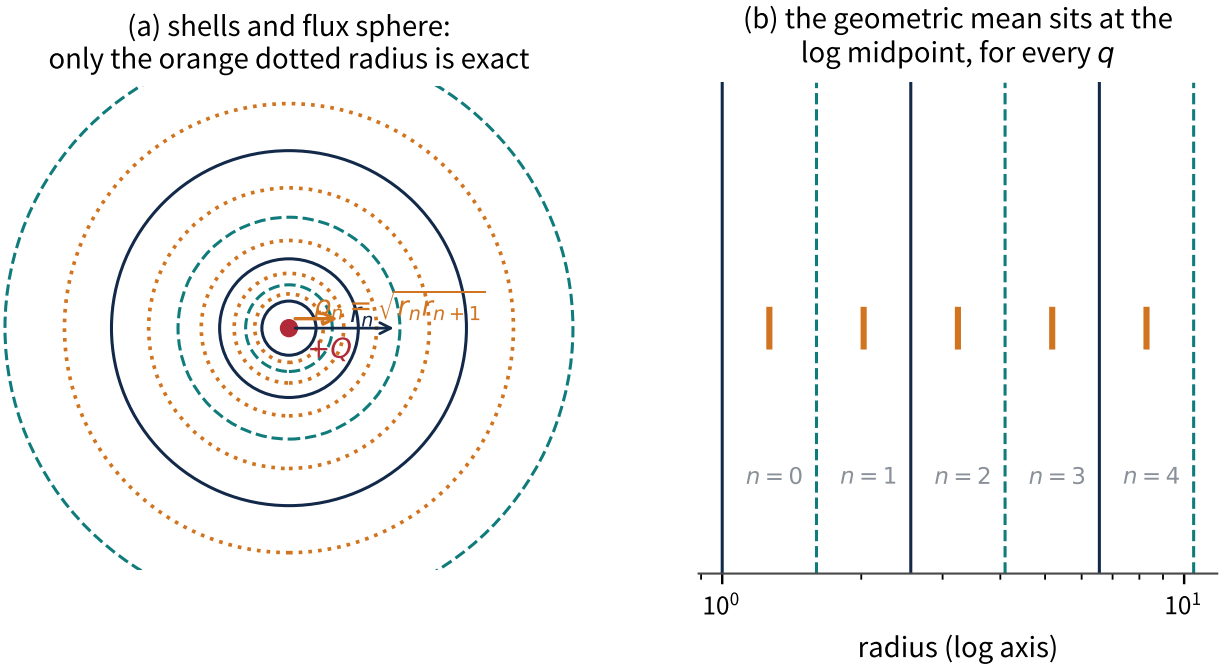


Figure 1: Shells and the flux sphere. The orange dotted radius is the only correct choice; it sits at the logarithmic midpoint of the two shells.

- Drop (i): the potential increments are themselves wrong and A_n acquires an extra factor λ^{-2} (or $1/q$, or $4q/(1+q)^2$). The exponent survives; the normalisation does not — and the normalisation is what is being measured.
- Drop (ii): $A_n = r_n^{p-d+2}\Omega_d^{-1}(q_n-1)/(q_n(1-q_n^{-p}))$ and $A_{n+1}/A_n = q^{p-d+2}$ is no longer exact. For $r_n = n^{1.3}$ and $p = 2$ the measured ratio departs from q_{eff}^{p-1} by up to 12%.
- In $d \neq 3$ the exponent is $p-d+2$, not $p-1$; marginality $p = d-2$ gives exponent zero in every dimension. Equation (7) is therefore q^{p-d+2}, and q^{p-1} is the three-dimensional special case: for $d = 4$, $p = 1$ the two read 0.707 and 1, a 41% difference. They cannot both be right.

Zero anomalous dimension is not triviality

A conserved current has zero anomalous dimension in an interacting theory: the electromagnetic current of QCD has $\gamma = 0$ by the Ward identity, and QCD is not free; the energy–momentum tensor of an interacting conformal field theory is the same story. What is claimed here is the scaling of **this flux observable**, not that the theory sits at a Gaussian fixed point. Statements about the whole theory would need all anomalous dimensions, not one.

Table 1: Independent audit: fitted log-log slope of A_n versus r_n.

d	p	fitted slope	predicted $p-d+2$
3	0.5	-0.500000	$-1/2$
3	1.0	0.000000	0
3	2.0	1.000000	1
4	1.0	-1.000000	-1
4	2.0	0.000000	0
5	3.0	0.000000	0

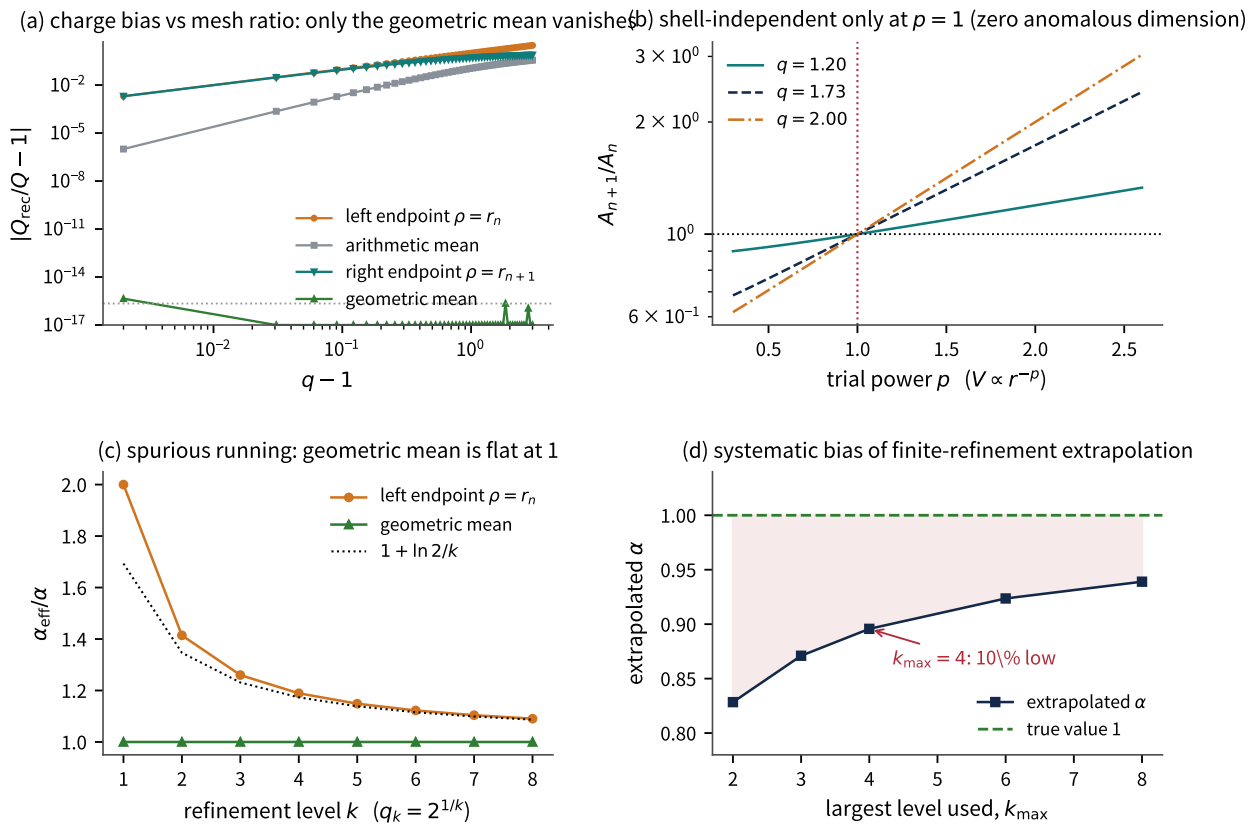


Figure 2: (a) charge bias for four estimators; (b) shell ratio q^{p-1} in $d = 3$, unity only at $p = 1$; (c) spurious running against $1 + \ln 2/k$; (d) extrapolation bias.

4 Two flows: refinement and renormalisation group

The *refinement flow* changes the description — mesh and estimator — at fixed physics; no degree of freedom is integrated out. The *RG flow* changes the theory: the scale μ, with a layer of modes removed. The only relation between them is

$$\beta_{\text{refine}} \equiv \left.\frac{\partial \alpha_k}{\partial \ln q_k}\right|_{\text{estimator}} = 0, \qquad \beta_{\text{phys}} \equiv \frac{\partial \alpha(\mu)}{\partial \ln \mu} \neq 0.$$

These are not two measurements of one quantity: the first is a property of the discretisation, the second a property of the theory. Any finite-refinement study that reports a non-zero β from a static flux without an estimator-error model has measured β_{refine} and called it β_{phys}.

One corollary: the two flows do not commute. Refining and then measuring differs from measuring and then refining by the estimator bias, which vanishes identically under the radius condition. The commutator is therefore a computable quantity, not a conceptual difficulty.

5 Consequence I: $\beta_{\text{refine}} = 0$ and the closed-form bias

With the geometric-mean radius the measured charge equals the continuum charge level by level, so $\beta_{\text{refine}} \equiv 0$ as an identity. For $q_k = 2^{1/k}$,

$$\alpha_{\text{eff}}/\alpha = \begin{cases} q_k, & \text{left endpoint} \\ 4q_k/(1+q_k)^2, & \text{arithmetic mean} \\ 1, & \text{geometric mean} \end{cases}$$

with $q_k = 1 + \ln 2/k + (\ln 2)^2/(2k^2) + \cdots$ and $4q_k/(1+q_k)^2 = 1 - (\ln 2)^2/(4k^2) + O(k^{-4})$ (the cubic term cancels exactly; the fitted exponent of the remainder is -4.00).

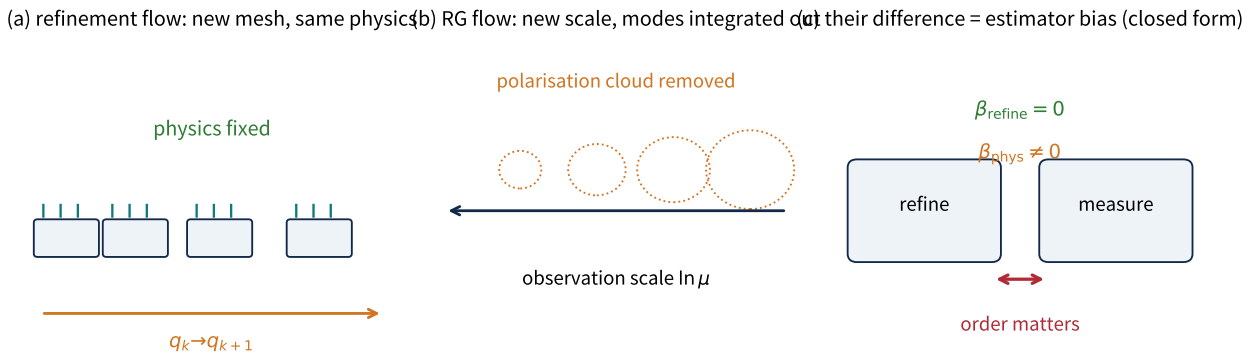


Figure 3: (a) refinement flow; (b) RG flow; (c) their difference is the closed-form estimator bias, while $\beta_{\text{refine}} = 0$.

Fitting $\alpha(1+c/k)$ to the first $k_{\max}$ levels gives $0.8284\,(2)$, $0.8710\,(3)$, $0.8957\,(4)$, $0.9236\,(6)$, $0.9390\,(8)$: 10.4% **low at** $k_{\max} = 4$, three orders of magnitude above the 10^{-4} precision usually claimed.

6 Consequence II: the scoped no-go

Theorem 2

Assume (H1) the charge is measured as a **static flux**; (H2) the estimator is a function of the shell geometry alone, with no fitting to dynamics; (H3) the field is Coulomb-like at the measurement scale. Then the anomalous dimension vanishes, the refinement fixed point is Gaussian, and **the coupling value is not selected by refinement invariance**.

This must not be read as a blanket prohibition. The correct statement is that **refinement invariance alone cannot determine the value of a non-Gaussian fixed-point coupling**. A specific number requires either a computed anomalous dimension or an independent quantisation condition — and in the second case the condition, not the geometry, does the work.

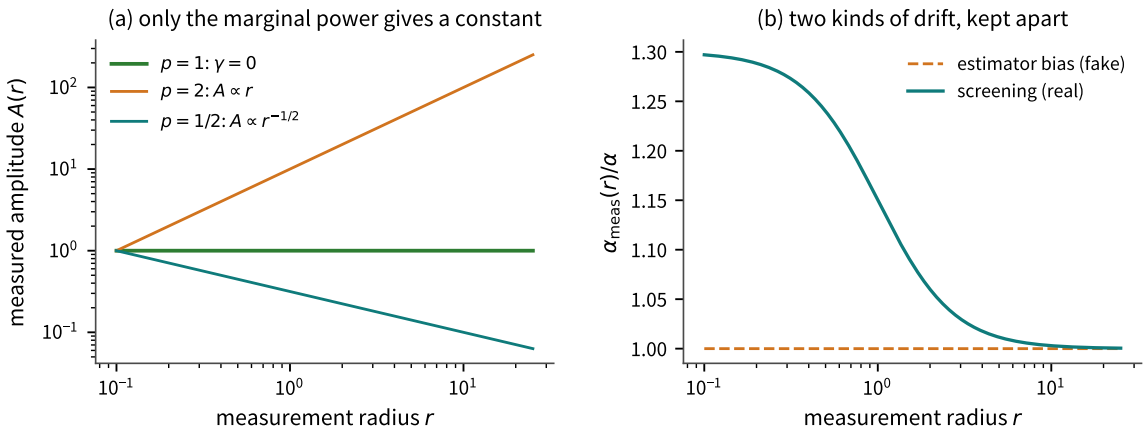


Figure 4: (a) only the marginal power is radius-independent; (b) estimator bias (fake) and screening (real) must be kept apart.

7 Non-Abelian: conditions for the Cartan extension

Lemma 2 (Cartan flux conservation)**.** *Let G be compact with maximal torus T of rank r, and E_i^a the Cartan electric field on link i. For any region S and any gauge-field configuration,*

$$\sum_{\text{links crossing } \partial S} E_i^a = Q_{\text{enc}}^a$$

exactly, configuration by configuration.

Proof. The Gauss constraint is $(D_i E_i)^a = \rho^a$ with $D_i E_i = \partial_i E_i + f^{abc} A_i^b E_i^c$. For a, b both Cartan, ${f^{ab}}_c = 0$ because T is Abelian, so on Cartan components $D_i E_i = \partial_i E_i$; summing over S telescopes by the lattice divergence theorem.

Two levels must be kept apart; conflating them is the most likely misuse of this section. **Level 1 (constraint, proved):** the lemma guarantees conservation of the Cartan flux and entails nothing about a coupling constant, an anomalous dimension or a fixed point. **Level 2 (coupling, conditional):** passing from a conserved flux to a coupling requires specifying the operator, the scheme, the configurations and the phase. Only then does Theorem 1 apply to that Cartan component. Hence three checks, not assumptions:

(C1) the observable is the static Cartan flux — not a gauge-dependent quantity, not an off-diagonal correlator;

(C2) no dynamical matter carrying Cartan charge inside the measurement volume (otherwise the enclosed charge is screened and Q_{enc} is not the charge of interest);

(C3) the measurement scale lies in the Coulomb phase of the Cartan sector (in a confined phase the flux is still conserved, but $p \neq d-2$ and the physical reading changes).

Under (C1)–(C3) together with a stated scheme, Theorems 1 and Lemma 1 hold componentwise and **the radius condition is unchanged** — it is a statement about the measure and carries no group-theoretic input, which is exactly why it transfers. What does not transfer is scale-freedom: off-diagonal modes and matter screen the Cartan charge, p departs from $d-2$, and q^{p-d+2} then measures a genuine anomalous dimension.

8 Lattices without spherical shells

(i) Exact route. The discrete flux through the boundary of a region is, by the divergence theorem, identically the enclosed charge. On a 96^3 lattice (with the neutralising background $|S|/L^3$ restored) this gives 1.000000000000 for $n = 2, 3, 5, 8, 12$. Exact, but it contains no comparison with $1/r$.

(ii) Radial projection. Project onto shells and use the radius of Theorem 1. The shell-decomposition error survives: the same data with a continuum-style estimator gives 0.900 ± 0.021 over $R = 8$–15, a 10% bias that decreases only as the sphere is resolved.

Arbitrary cells. The real content is that the flux weight must equal the exact potential difference, $w_C = \Omega_d^{-1} \Delta(r^{2-d})|_C$ cell by cell. *No spheres needed — weights are.*

9 The null test

10 Relation to existing methods, and what is new

- **Finite differences on graded meshes.** Second-order accuracy of a symmetric (midpoint) scheme on a non-uniform mesh is classical; exponential meshes are standard in atomic structure codes. Theorem 1 is the *exact* version: on a geometric mesh the q-symmetric point is $r_n\sqrt{q}$, and for the $1/r$ field the error there is not second order but *identically zero*, at finite q. In q-calculus language, the flux must be sampled at the q-symmetric point.

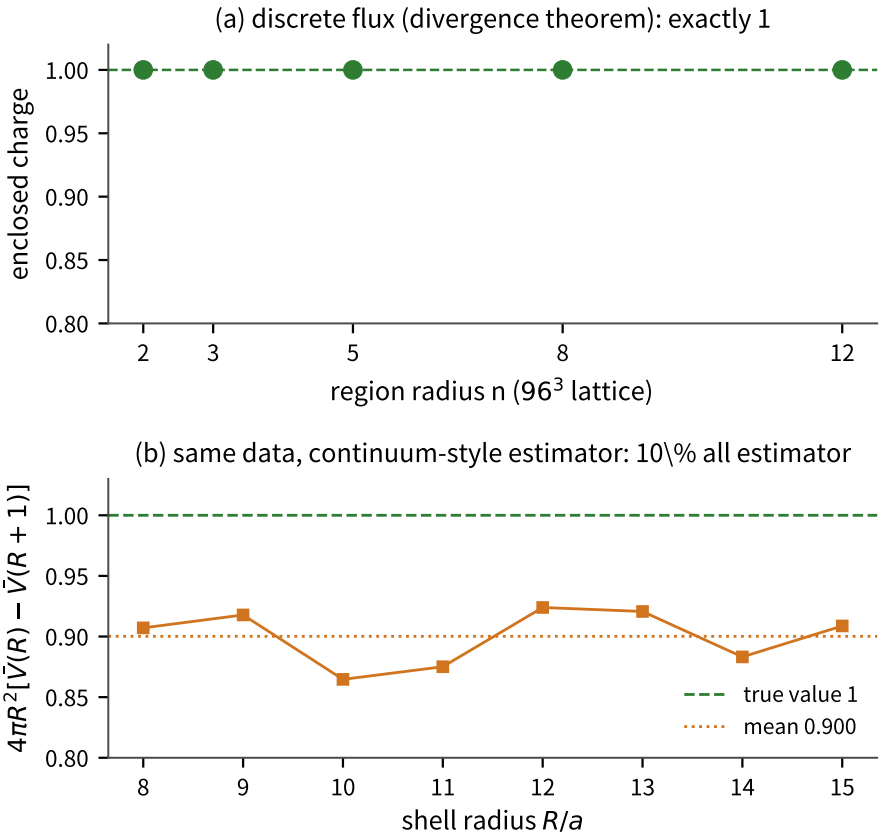


Figure 5: (a) divergence-theorem flux: exactly 1; (b) continuum-style estimator: 0.900, the bias is estimator plus lattice geometry.

- **Discrete Gauss law.** Fundamental since Wilson; conservation of Cartan flux is standard lore in the Abelian-projection literature. What is not usually exploited is that on a *radial shell* decomposition the constraint reproduces the continuum $1/r$ exactly, because homogeneous lattices carry no distinguished radial measure.
- **Flux and static-potential estimators.** Lattice fits of $V = -C/r + \sigma r + a + O(a^2)$ absorb the estimator bias into fitted terms; Symanzik improvement removes the $O(a^2)$ of the *action*. This paper acts on a third axis: it removes the $O(a^0)$ estimator error from the *measurement* at finite a without touching the action. The three are complementary and not substitutes.

Statement of originality (strict)

The telescoping identity is elementary and is not claimed. What is claimed: (i) the if-and-only-if characterisation of exact estimators on arbitrary meshes; (ii) the equivalence exactness ⇔ marginality ⇔ zero anomalous dimension; (iii) the closed-form biases of the natural estimators and their size at realistic refinement depths; (iv) the scoped no-go for couplings obtained from refinement invariance. The Cartan lemma is standard; its role here is an applicability condition.

11 Three follow-ups

(a) Post-hoc audit. Any study quoting a coupling from a sequence q_k can be audited without re-running: the closed form of $\alpha_{\text{eff}}/\alpha$ gives the bias of the estimator it used.

(b) Dynamical matter and non-Cartan components. (C2)–(C3) are where dynamics enters. The natural object is the measured exponent p; $p - (d-2)$ is the anomalous dimension of the sector. A non-Cartan charge has no conserved flux of the type $\sum E = Q$, so the criterion is expected to fail there —

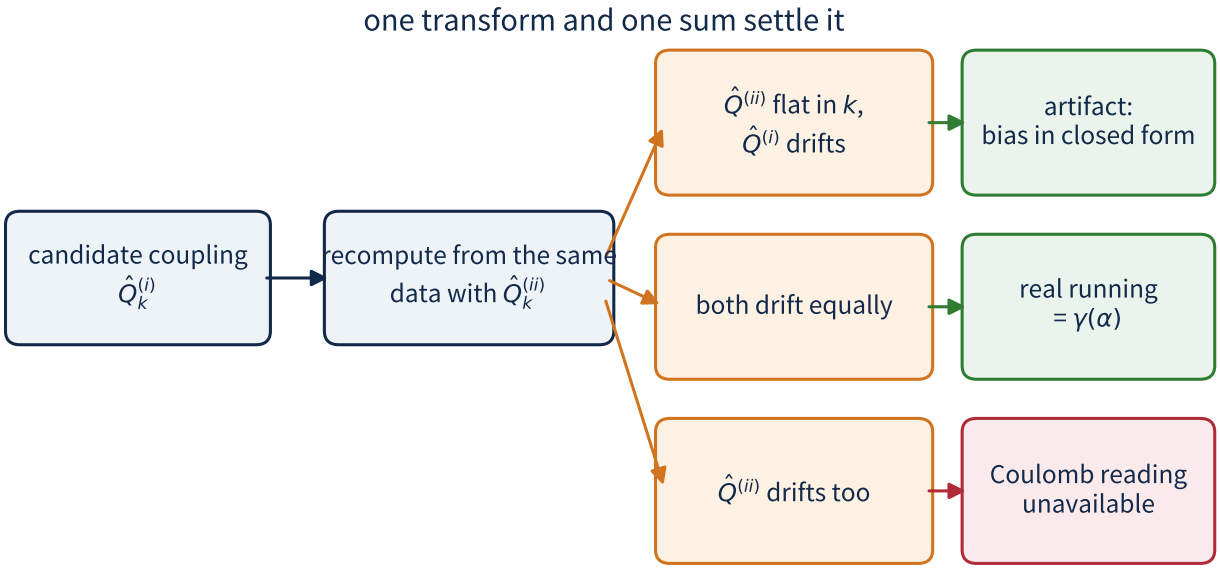


Figure 6: Change only the estimator on the same data: a flat $\hat{Q}^{(ii)}$ with a drifting $\hat{Q}^{(i)}$ means artifact; equal drift means real running; a drifting $\hat{Q}^{(ii)}$ means the Coulomb reading is unavailable.

and the failure is itself informative.

(c) Weights on arbitrary cell complexes. The conjecture is that on a Voronoi–Delaunay complex the weight making the criterion exact is the harmonic measure of the cell boundary. If true, the criterion applies to simplicial and hypercubic lattices on the same footing, and the last spherical assumption disappears.

12 Code and verification

Table 2: Verification suite. Every number in the paper comes from these scripts; logs are included.

script	content	time
`t1_exact_gauss.py`	Theorem 1: four estimators, λ family, random perturbations	0.09 s
`t3_lattice_green.py`	96^3 divergence theorem; continuum estimator; arbitrary meshes	0.23 s
`t4_running.py`	q-universality; spurious running; extrapolation bias	0.09 s
`t5_marginality.py`	$A_{n+1}/A_n = q^{p-d+2}$ for $p = 0.5$–3, $d = 3, 4, 5$	0.08 s
`t6_exponents.py`	exponent audit: $d = 3, 4, 5$; variable q; the two closed forms	0.10 s
`make_figures.py`	paper figures 1 and 2	1.3 s
`make_guide_figures.py`	all figures here (both languages)	7 s

Key numbers: criterion residual 5.48×10^{-13}; closed forms agree to 0–10^{-16}; random perturbations $[1.12, 1.72]\varepsilon$; fitted exponents agree with prediction to $\leq 3 \times 10^{-16}$; variable-q departure up to 12%; extrapolation at $k_{\max} = 4$ gives 0.8957 (10.4% low); 96^3 flux 1.000000000000.

13 Reproducing

Code and all logs: `https://github.com/csoftxyz/RIA_EISA/tree/main/Coulomb` (directories `code/` and `logs/`).

```
cd refinement_flow
```

```
sh code/run_all.sh
pdflatex main_prd.tex && pdflatex main_prd.tex
xelatex guide/guide_zh.tex
xelatex guide/guide_en.tex
```

If you remember one line

The "refinement running" of the Coulomb coupling is a measurement error that can be removed exactly; what remains after removing it is not a number but a question about anomalous dimensions.